

CSE 211 – Turing Machine Problems

Compact Finite-Control Transition Functions

Notation

Every transition is written directly in exam form:

$$\delta(\text{state, scanned symbol}) = (\text{next state, written symbol, move}).$$

For 2-track machines, a tape cell is $[m, x]$, where $m \in \{B, *\}$ is the marker track and x is the data track. Finite-control storage is written inside the state, e.g. $[q, a]$. The shorthand $0/1$ means either 0 or 1 when both cases have exactly the same action. Likewise $B/*$ means either marker value when the action is identical. Halting states have no outgoing transitions.

Contents

1 2023–24 Q9 – Binary Equality Comparator

Problem statement

The tape initially contains w_1cw_2c , where w_1, w_2 are equal-length binary strings. Construct a TM that appends 0 after the second c when $w_1 = w_2$, and appends 1 otherwise. A 2-track tape and finite-control storage may be used.

State meanings. $[q_0, B]$: choose the next unchecked bit of w_1 ; $[q_1, a]$: carry bit $a \in \{0, 1\}$ across w_1 ; $[q_2, a]$: compare the stored bit with the next unchecked bit of w_2 ; $[q_r, B]$: return toward the input start; $[q_{w0}, B]$: append 0; $[q_{w1}, B]$: append 1; $[q_h, B]$: halt.

Complete transition function

$$\begin{aligned}
 \delta([q_0, B], [*, 0/1]) &= ([q_0, B], [*, 0/1], R) \\
 \delta([q_0, B], [B, a]) &= ([q_1, a], [*, a], R), \quad a \in \{0, 1\} \\
 \delta([q_0, B], [B, c]) &= ([q_{w0}, B], [B, c], R) \\
 \delta([q_1, a], [B, 0/1]) &= ([q_1, a], [B, 0/1], R) \\
 \delta([q_1, a], [B, c]) &= ([q_2, a], [B, c], R) \\
 \delta([q_2, a], [*, 0/1]) &= ([q_2, a], [*, 0/1], R) \\
 \delta([q_2, a], [B, a]) &= ([q_r, B], [*, a], L) \\
 \delta([q_2, a], [B, \bar{a}]) &= ([q_{w1}, B], [*, \bar{a}], R) \\
 \delta([q_r, B], [B/*, 0/1]) &= ([q_r, B], [B/*, 0/1], L) \\
 \delta([q_r, B], [B, c]) &= ([q_r, B], [B, c], L) \\
 \delta([q_r, B], [B, B]) &= ([q_0, B], [B, B], R) \\
 \delta([q_{w0}, B], [B/*, 0/1]) &= ([q_{w0}, B], [B/*, 0/1], R) \\
 \delta([q_{w0}, B], [B, c]) &= ([q_{w0}, B], [B, c], R) \\
 \delta([q_{w0}, B], [B, B]) &= ([q_h, B], [B, 0], S) \\
 \delta([q_{w1}, B], [B/*, 0/1]) &= ([q_{w1}, B], [B/*, 0/1], R) \\
 \delta([q_{w1}, B], [B, c]) &= ([q_{w1}, B], [B, c], R) \\
 \delta([q_{w1}, B], [B, B]) &= ([q_h, B], [B, 1], S)
 \end{aligned}$$

2 2022–23 Q10 – Logical NOT Appender

Problem statement

The tape initially contains wc , where w is binary. Construct a TM that appends the bitwise NOT of w after c and then halts.

State meanings. $[q_0, B]$: choose the next unchecked input bit; $[q_s, \bar{a}]$: carry the complemented bit $\bar{a}$ in finite control; $[q_r, B]$: return toward the input start; $[q_h, B]$: halt.

Complete transition function

$$\begin{aligned}
 \delta([q_0, B], [*, 0/1]) &= ([q_0, B], [*, 0/1], R) \\
 \delta([q_0, B], [B, a]) &= ([q_s, \bar{a}], [*, a], R), \quad a \in \{0, 1\} \\
 \delta([q_0, B], [B, c]) &= ([q_h, B], [B, c], S) \\
 \delta([q_s, \bar{a}], [B/*, 0/1]) &= ([q_s, \bar{a}], [B/*, 0/1], R) \\
 \delta([q_s, \bar{a}], [B, c]) &= ([q_s, \bar{a}], [B, c], R) \\
 \delta([q_s, \bar{a}], [B, B]) &= ([q_r, B], [B, \bar{a}], L) \\
 \delta([q_r, B], [B/*, 0/1]) &= ([q_r, B], [B/*, 0/1], L) \\
 \delta([q_r, B], [B, c]) &= ([q_r, B], [B, c], L) \\
 \delta([q_r, B], [B, B]) &= ([q_0, B], [B, B], R)
 \end{aligned}$$

3 2021–22 Q9 – Signed-Magnitude to Two’s Complement

Problem statement

The tape contains a signed-magnitude binary word with the sign bit first. If the sign bit is 0, leave the word unchanged. If it is 1, convert the magnitude to two’s complement by scanning from the LSB side: keep trailing 0s, keep the first 1, then invert the remaining magnitude bits. The final tape must contain only the converted binary word.

State meanings. q_0 : inspect the sign bit; q_s : scan to the blank after the word; q_c : preserve trailing 0s; q_f : invert the remaining magnitude bits; q_h : halt. S temporarily marks the negative sign bit.

Complete transition function

$$\begin{aligned}\delta(q_0, 0) &= (q_h, 0, S) \\ \delta(q_0, 1) &= (q_s, S, R) \\ \delta(q_s, 0/1) &= (q_s, 0/1, R) \\ \delta(q_s, B) &= (q_c, B, L) \\ \delta(q_c, 0) &= (q_c, 0, L) \\ \delta(q_c, 1) &= (q_f, 1, L) \\ \delta(q_c, S) &= (q_h, 1, S) \\ \delta(q_f, 0) &= (q_f, 1, L) \\ \delta(q_f, 1) &= (q_f, 0, L) \\ \delta(q_f, S) &= (q_h, 1, S)\end{aligned}$$

4 2019–20 Q9 – Logical AND Appender

Problem statement

The tape initially contains w_1cw_2c , where w_1, w_2 are equal-length binary strings. Construct a TM that appends $w_1 \wedge w_2$ after the second c and halts.

State meanings. $[q_0, B]$: choose the next unchecked bit a of w_1 ; $[q_1, a]$: carry a across w_1 ; $[q_2, a]$: read the corresponding bit b of w_2 ; $[q_o, d]$: carry result $d = a \wedge b$ to the output end; $[q_r, B]$: return toward the input start; $[q_h, B]$: halt.

Complete transition function

$$\begin{aligned}\delta([q_0, B], [*, 0/1]) &= ([q_0, B], [*, 0/1], R) \\ \delta([q_0, B], [B, a]) &= ([q_1, a], [*, a], R), \quad a \in \{0, 1\} \\ \delta([q_0, B], [B, c]) &= ([q_h, B], [B, c], S) \\ \delta([q_1, a], [B, 0/1]) &= ([q_1, a], [B, 0/1], R) \\ \delta([q_1, a], [B, c]) &= ([q_2, a], [B, c], R) \\ \delta([q_2, a], [*, 0/1]) &= ([q_2, a], [*, 0/1], R) \\ \delta([q_2, a], [B, b]) &= ([q_o, a \wedge b], [*, b], R), \quad a, b \in \{0, 1\} \\ \delta([q_o, d], [B/*, 0/1]) &= ([q_o, d], [B/*, 0/1], R) \\ \delta([q_o, d], [B, c]) &= ([q_o, d], [B, c], R) \\ \delta([q_o, d], [B, B]) &= ([q_r, B], [B, d], L) \\ \delta([q_r, B], [B/*, 0/1]) &= ([q_r, B], [B/*, 0/1], L) \\ \delta([q_r, B], [B, c]) &= ([q_r, B], [B, c], L) \\ \delta([q_r, B], [B, B]) &= ([q_0, B], [B, B], R)\end{aligned}$$

5 Jan 2020 Q9 – Logical XOR Appender

Problem statement

The tape initially contains w_1cw_2c , where w_1, w_2 are equal-length binary strings. Construct a TM that appends $w_1 \oplus w_2$ after the second c and halts.

State meanings. $[q_0, B]$: choose the next unchecked bit a of w_1 ; $[q_1, a]$: carry a across w_1 ; $[q_2, a]$: read the corresponding bit b of w_2 ; $[q_o, d]$: carry result $d = a \oplus b$ to the output end; $[q_r, B]$: return toward the input start; $[q_h, B]$: halt.

Complete transition function

$$\begin{aligned}
 \delta([q_0, B], [*, 0/1]) &= ([q_0, B], [*, 0/1], R) \\
 \delta([q_0, B], [B, a]) &= ([q_1, a], [*, a], R), \quad a \in \{0, 1\} \\
 \delta([q_0, B], [B, c]) &= ([q_h, B], [B, c], S) \\
 \delta([q_1, a], [B, 0/1]) &= ([q_1, a], [B, 0/1], R) \\
 \delta([q_1, a], [B, c]) &= ([q_2, a], [B, c], R) \\
 \delta([q_2, a], [*, 0/1]) &= ([q_2, a], [*, 0/1], R) \\
 \delta([q_2, a], [B, b]) &= ([q_o, a \oplus b], [*, b], R), \quad a, b \in \{0, 1\} \\
 \delta([q_o, d], [B/*, 0/1]) &= ([q_o, d], [B/*, 0/1], R) \\
 \delta([q_o, d], [B, c]) &= ([q_o, d], [B, c], R) \\
 \delta([q_o, d], [B, B]) &= ([q_r, B], [B, d], L) \\
 \delta([q_r, B], [B/*, 0/1]) &= ([q_r, B], [B/*, 0/1], L) \\
 \delta([q_r, B], [B, c]) &= ([q_r, B], [B, c], L) \\
 \delta([q_r, B], [B, B]) &= ([q_0, B], [B, B], R)
 \end{aligned}$$

6 2017–18 Q9 – Logical OR Appender

Problem statement

The tape initially contains $N_1\#N_2\#$, where N_1, N_2 are equal-length binary strings. Construct a TM that appends $N_1 \vee N_2$ after the second $\#$ and halts.

State meanings. $[q_0, B]$: choose the next unchecked bit a of N_1 ; $[q_1, a]$: carry a across N_1 ; $[q_2, a]$: read the corresponding bit b of N_2 ; $[q_o, d]$: carry result $d = a \vee b$ to the output end; $[q_r, B]$: return toward the input start; $[q_h, B]$: halt.

Complete transition function

$$\begin{aligned}
 \delta([q_0, B], [*, 0/1]) &= ([q_0, B], [*, 0/1], R) \\
 \delta([q_0, B], [B, a]) &= ([q_1, a], [*, a], R), \quad a \in \{0, 1\} \\
 \delta([q_0, B], [B, \#]) &= ([q_h, B], [B, \#], S) \\
 \delta([q_1, a], [B, 0/1]) &= ([q_1, a], [B, 0/1], R) \\
 \delta([q_1, a], [B, \#]) &= ([q_2, a], [B, \#], R) \\
 \delta([q_2, a], [*, 0/1]) &= ([q_2, a], [*, 0/1], R) \\
 \delta([q_2, a], [B, b]) &= ([q_o, a \vee b], [*, b], R), \quad a, b \in \{0, 1\} \\
 \delta([q_o, d], [B/*, 0/1]) &= ([q_o, d], [B/*, 0/1], R) \\
 \delta([q_o, d], [B, \#]) &= ([q_o, d], [B, \#], R) \\
 \delta([q_o, d], [B, B]) &= ([q_r, B], [B, d], L) \\
 \delta([q_r, B], [B/*, 0/1]) &= ([q_r, B], [B/*, 0/1], L) \\
 \delta([q_r, B], [B, \#]) &= ([q_r, B], [B, \#], L) \\
 \delta([q_r, B], [B, B]) &= ([q_0, B], [B, B], R)
 \end{aligned}$$

7 2016–17 Q5(b) – Binary Incrementer

Problem statement

The tape initially contains $\#N$, where N is a binary number and the head starts on $\#$. Construct a TM that replaces N by $N + 1$ and halts. The $\#$ may be replaced when overflow occurs.

State meanings. q_0 : pass the separator; q_s : scan across N ; q_a : ripple-carry addition; q_d : carry is finished; q_h : halt.

Complete transition function

$$\begin{aligned}\delta(q_0, \#) &= (q_s, \#, R) \\ \delta(q_s, 0/1) &= (q_s, 0/1, R) \\ \delta(q_s, B) &= (q_a, B, L) \\ \delta(q_a, 1) &= (q_a, 0, L) \\ \delta(q_a, 0) &= (q_d, 1, L) \\ \delta(q_a, \#) &= (q_h, 1, S) \\ \delta(q_d, 0/1) &= (q_d, 0/1, L) \\ \delta(q_d, \#) &= (q_h, \#, R)\end{aligned}$$

8 2016–17 Q6(c) – Equal Number of 0s and 1s

Problem statement

Construct a TM that decides the set of all binary strings containing an equal number of 0s and 1s, in any order.

State meanings. q_0 : choose the next unmarked symbol; q_1 : after marking 0, search for 1; q'_0 : after marking 1, search for 0; q_r : return toward the input start; q_a : accept; q_x : reject. Marked 0s are X and marked 1s are Y .

Complete transition function

$$\begin{aligned}\delta(q_0, X/Y) &= (q_0, X/Y, R) \\ \delta(q_0, 0) &= (q_1, X, R) \\ \delta(q_0, 1) &= (q'_0, Y, R) \\ \delta(q_0, B) &= (q_a, B, S) \\ \delta(q_1, 0/X/Y) &= (q_1, 0/X/Y, R) \\ \delta(q_1, 1) &= (q_r, Y, L) \\ \delta(q_1, B) &= (q_x, B, S) \\ \delta(q'_0, 1/X/Y) &= (q'_0, 1/X/Y, R) \\ \delta(q'_0, 0) &= (q_r, X, L) \\ \delta(q'_0, B) &= (q_x, B, S) \\ \delta(q_r, 0/1/X/Y) &= (q_r, 0/1/X/Y, L) \\ \delta(q_r, B) &= (q_0, B, R)\end{aligned}$$

9 2015–16 Q7(a) – $\{a^n b^n c^n \mid n \geq 1\}$

Problem statement

Construct a TM that decides $L = \{a^n b^n c^n \mid n \geq 1\}$.

State meanings. q_0 : choose an unmarked a ; q_b : match one b ; q_c : match one c ; q_r : return toward the input start; q_v : verify completion; q_a : accept; q_x : reject. Markers are X, Y, Z .

Complete transition function

$$\begin{aligned}
\delta(q_0, X) &= (q_0, X, R) \\
\delta(q_0, a) &= (q_b, X, R) \\
\delta(q_0, Y/Z) &= (q_v, Y/Z, S) \\
\delta(q_0, b/c/B) &= (q_x, b/c/B, S) \\
\delta(q_b, a/Y) &= (q_b, a/Y, R) \\
\delta(q_b, b) &= (q_c, Y, R) \\
\delta(q_b, c/Z/B) &= (q_x, c/Z/B, S) \\
\delta(q_c, b/Z) &= (q_c, b/Z, R) \\
\delta(q_c, c) &= (q_r, Z, L) \\
\delta(q_c, a/Y/B) &= (q_x, a/Y/B, S) \\
\delta(q_r, a/b/c/X/Y/Z) &= (q_r, a/b/c/X/Y/Z, L) \\
\delta(q_r, B) &= (q_0, B, R) \\
\delta(q_v, Y/Z) &= (q_v, Y/Z, R) \\
\delta(q_v, B) &= (q_a, B, S) \\
\delta(q_v, a/b/c/X) &= (q_x, a/b/c/X, S)
\end{aligned}$$

10 2015–16 Q8(a) – Monus Variant $\max(b - a, 0)$

Problem statement

The tape contains $1^a 0 1^b$. Construct a TM that computes $\max(b - a, 0)$ and leaves only the unary output $1^{\max(b-a, 0)}$ on the tape.

State meanings. q_0 : cancel one 1 from the first block; q_s : cross to the separator; q_b : cancel one 1 from the second block; q_r : return toward the input start; q_{cl}, q_D, q_Y : cleanup when the first block is exhausted; q_z : erase everything when the second block is exhausted first; q_h : halt. X, Y are cancellation marks and D is a temporary separator mark.

Complete transition function

$$\begin{aligned}
\delta(q_0, X) &= (q_0, X, R) \\
\delta(q_0, 1) &= (q_s, X, R) \\
\delta(q_0, 0) &= (q_{cl}, D, L) \\
\delta(q_s, 1/X) &= (q_s, 1/X, R) \\
\delta(q_s, 0) &= (q_b, 0, R) \\
\delta(q_b, Y) &= (q_b, Y, R) \\
\delta(q_b, 1) &= (q_r, Y, L) \\
\delta(q_b, B) &= (q_z, B, L) \\
\delta(q_r, 1/0/X/Y) &= (q_r, 1/0/X/Y, L) \\
\delta(q_r, B) &= (q_0, B, R) \\
\delta(q_{cl}, X) &= (q_{cl}, B, L) \\
\delta(q_{cl}, B) &= (q_D, B, R) \\
\delta(q_D, B) &= (q_D, B, R) \\
\delta(q_D, D) &= (q_Y, B, R) \\
\delta(q_Y, Y) &= (q_Y, B, R) \\
\delta(q_Y, 1) &= (q_h, 1, S) \\
\delta(q_Y, B) &= (q_h, B, S) \\
\delta(q_z, 1/0/X/Y) &= (q_z, B, L) \\
\delta(q_z, B) &= (q_h, B, S)
\end{aligned}$$

11 2013–14 Q2(a) – $\{0^N 1^N 0^N \mid N \geq 0\}$

Problem statement

Construct a TM that decides $L = \{0^N 1^N 0^N \mid N \geq 0\}$.

State meanings. q_0 : choose a 0 from the first block; q_1 : match a 1 from the second block; q_2 : match a 0 from the third block; q_r : return toward the input start; q_v : verify completion; q_a : accept; q_x : reject. Markers are X, Y, Z .

Complete transition function

$$\begin{aligned}
 \delta(q_0, X) &= (q_0, X, R) \\
 \delta(q_0, 0) &= (q_1, X, R) \\
 \delta(q_0, Y/Z) &= (q_v, Y/Z, S) \\
 \delta(q_0, B) &= (q_a, B, S) \\
 \delta(q_0, 1) &= (q_x, 1, S) \\
 \delta(q_1, 0/Y) &= (q_1, 0/Y, R) \\
 \delta(q_1, 1) &= (q_2, Y, R) \\
 \delta(q_1, Z/B) &= (q_x, Z/B, S) \\
 \delta(q_2, 1/Z) &= (q_2, 1/Z, R) \\
 \delta(q_2, 0) &= (q_r, Z, L) \\
 \delta(q_2, B) &= (q_x, B, S) \\
 \delta(q_r, 0/1/X/Y/Z) &= (q_r, 0/1/X/Y/Z, L) \\
 \delta(q_r, B) &= (q_0, B, R) \\
 \delta(q_v, Y/Z) &= (q_v, Y/Z, R) \\
 \delta(q_v, B) &= (q_a, B, S) \\
 \delta(q_v, 0/1/X) &= (q_x, 0/1/X, S)
 \end{aligned}$$

12 2012–13 Q3(b) – $\{w - w - w\}$

Problem statement

Let $\Sigma = \{-, a, b\}$ and $L = \{w - w - w \mid w \in \{a, b\}^*\}$. Construct a TM that decides L .

State meanings. $[q_0, B]$: choose a symbol $\sigma \in \{a, b\}$ from the first block; $[q_1, \sigma]$: carry σ to block two; $[q_2, \sigma]$: match σ in block two; $[q_3, \sigma]$: carry σ to block three; $[q_4, \sigma]$: match σ in block three; $[q_r, B]$: return toward the input start; q_v, q_{v3} : verify completion; q_a : accept; q_x : reject. Define markers $M_1(a) = A, M_1(b) = B_1, M_2(a) = C, M_2(b) = D, M_3(a) = E, M_3(b) = F$.

Complete transition function

$$\begin{aligned}
 \delta([q_0, B], A/B_1) &= ([q_0, B], A/B_1, R) \\
 \delta([q_0, B], \sigma) &= ([q_1, \sigma], M_1(\sigma), R), \quad \sigma \in \{a, b\} \\
 \delta([q_0, B], -) &= ([q_v, B], -, R) \\
 \delta([q_1, \sigma], a/b/A/B_1) &= ([q_1, \sigma], a/b/A/B_1, R) \\
 \delta([q_1, \sigma], -) &= ([q_2, \sigma], -, R) \\
 \delta([q_2, \sigma], C/D) &= ([q_2, \sigma], C/D, R) \\
 \delta([q_2, \sigma], \sigma) &= ([q_3, \sigma], M_2(\sigma), R) \\
 \delta([q_2, a], b/-) &= ([q_x, B], b/-, S) \\
 \delta([q_2, b], a/-) &= ([q_x, B], a/-, S) \\
 \delta([q_3, \sigma], a/b/C/D) &= ([q_3, \sigma], a/b/C/D, R) \\
 \delta([q_3, \sigma], -) &= ([q_4, \sigma], -, R) \\
 \delta([q_4, \sigma], E/F) &= ([q_4, \sigma], E/F, R)
 \end{aligned}$$

$$\begin{aligned}
\delta([q_4, \sigma], \sigma) &= ([q_r, B], M_3(\sigma), L) \\
\delta([q_4, a], b/B) &= ([q_x, B], b/B, S) \\
\delta([q_4, b], a/B) &= ([q_x, B], a/B, S) \\
\delta([q_r, B], a/b/ - /A/B_1/C/D/E/F) &= ([q_r, B], a/b/ - /A/B_1/C/D/E/F, L) \\
\delta([q_r, B], B) &= ([q_0, B], B, R) \\
\delta([q_v, B], C/D) &= ([q_v, B], C/D, R) \\
\delta([q_v, B], -) &= ([q_{v3}, B], -, R) \\
\delta([q_v, B], a/b) &= ([q_x, B], a/b, S) \\
\delta([q_{v3}, B], E/F) &= ([q_{v3}, B], E/F, R) \\
\delta([q_{v3}, B], B) &= ([q_a, B], B, S) \\
\delta([q_{v3}, B], a/b/-) &= ([q_x, B], a/b/-, S)
\end{aligned}$$

13 2012–13 Q3(c) – Parity Rewrite

Problem statement

The input is $\#w$, $w \in \{0, 1\}^*$, $n = |w|$. If n is even, replace w by 0^n . If n is odd, replace w by 1^n . Then accept.

State meanings. q_s : pass $\#$; q_e : even parity; q_o : odd parity; q_{w0} : rewrite bits as 0; q_{w1} : rewrite bits as 1; q_a : accept.

Complete transition function

$$\begin{aligned}
\delta(q_s, \#) &= (q_e, \#, R) \\
\delta(q_e, 0/1) &= (q_o, 0/1, R) \\
\delta(q_e, B) &= (q_{w0}, B, L) \\
\delta(q_o, 0/1) &= (q_e, 0/1, R) \\
\delta(q_o, B) &= (q_{w1}, B, L) \\
\delta(q_{w0}, 0/1) &= (q_{w0}, 0, L) \\
\delta(q_{w0}, \#) &= (q_a, \#, R) \\
\delta(q_{w1}, 0/1) &= (q_{w1}, 1, L) \\
\delta(q_{w1}, \#) &= (q_a, \#, R)
\end{aligned}$$

14 2011–12 Q1(a) – Modified Unary Subtraction

Problem statement

Numbers use the modified unary encoding $N \mapsto 1^{N+1}$. The input is $1^{x+1}01^{y+1}$ with $x \geq y$. Construct a TM that leaves the modified-unary encoding of $x - y$.

State meanings. q_0 : select one 1 from the first block; q_s : cross the separator; q_y : locate an uncanceled right-block 1; q_p : test whether that 1 is the final sentinel; q_m : mark a normal cancellation; q_r : return toward the input start; q_f, q_c : final cleanup; q_h : halt.

Complete transition function

$$\begin{aligned}
\delta(q_0, X) &= (q_0, X, R) \\
\delta(q_0, 1) &= (q_s, X, R) \\
\delta(q_s, 1) &= (q_s, 1, R) \\
\delta(q_s, 0) &= (q_y, 0, R) \\
\delta(q_y, Y) &= (q_y, Y, R) \\
\delta(q_y, 1) &= (q_p, 1, R)
\end{aligned}$$

$\delta(q_p, 1) = (q_m, 1, L)$
 $\delta(q_p, B) = (q_f, B, L)$
 $\delta(q_m, 1) = (q_r, Y, L)$
 $\delta(q_r, 1/0/X/Y) = (q_r, 1/0/X/Y, L)$
 $\delta(q_r, B) = (q_0, B, R)$
 $\delta(q_f, 1) = (q_f, 1, L)$
 $\delta(q_f, Y/0) = (q_f, B, L)$
 $\delta(q_f, X) = (q_c, 1, L)$
 $\delta(q_c, X) = (q_c, B, L)$
 $\delta(q_c, B) = (q_h, B, S)$

15 2011–12 Q2(a) – 2-Tape Decider for $\{wcw^R\}$

Problem statement

Construct a 2-tape TM that decides $L = \{wcw^R \mid w \in \{a, b\}^*\}$.

State meanings. q_c : copy w from Tape 1 to Tape 2; q_m : compare Tape 1 after c against Tape 2 in reverse order; q_a : accept; q_x : reject. A 2-tape transition is $\delta(q, (x_1, x_2)) = (p, (y_1, y_2), (D_1, D_2))$.

Complete transition function

$\delta(q_c, (a, B)) = (q_c, (a, a), (R, R))$
 $\delta(q_c, (b, B)) = (q_c, (b, b), (R, R))$
 $\delta(q_c, (c, B)) = (q_m, (c, B), (R, L))$
 $\delta(q_m, (a, a)) = (q_m, (a, a), (R, L))$
 $\delta(q_m, (b, b)) = (q_m, (b, b), (R, L))$
 $\delta(q_m, (a, b)) = (q_x, (a, b), (S, S))$
 $\delta(q_m, (b, a)) = (q_x, (b, a), (S, S))$
 $\delta(q_m, (B, B)) = (q_a, (B, B), (S, S))$
 $\delta(q_m, (B, a/b)) = (q_x, (B, a/b), (S, S))$
 $\delta(q_m, (a/b, B)) = (q_x, (a/b, B), (S, S))$

16 2009–10 Q12(a) – Scan Until Two Consecutive as

Problem statement

Construct a TM over $\{a, b, B\}$ that scans to the right and halts when it encounters two consecutive as .

State meanings. q_0 : previous symbol was not a ; q_1 : previous scanned symbol was a ; q_h : two consecutive as found; q_e : input ended first.

Complete transition function

$\delta(q_0, a) = (q_1, a, R)$
 $\delta(q_0, b) = (q_0, b, R)$
 $\delta(q_0, B) = (q_e, B, S)$
 $\delta(q_1, a) = (q_h, a, S)$
 $\delta(q_1, b) = (q_0, b, R)$
 $\delta(q_1, B) = (q_e, B, S)$

17 2007–08 Q7 – $\{0^n 1^n \mid n \geq 1\}$

Problem statement

Construct a TM that decides $L = \{0^n 1^n \mid n \geq 1\}$.

State meanings. q_0 : choose an unmarked 0; q_1 : search for a matching unmarked 1; q_2 : return toward the input start; q_v : verify only Y markers remain; q_a : accept; q_x : reject. X marks matched 0s and Y marks matched 1s.

Complete transition function

$$\delta(q_0, X) = (q_0, X, R)$$

$$\delta(q_0, 0) = (q_1, X, R)$$

$$\delta(q_0, Y) = (q_v, Y, R)$$

$$\delta(q_0, 1/B) = (q_x, 1/B, S)$$

$$\delta(q_1, 0/Y) = (q_1, 0/Y, R)$$

$$\delta(q_1, 1) = (q_2, Y, L)$$

$$\delta(q_1, B) = (q_x, B, S)$$

$$\delta(q_2, 0/1/Y) = (q_2, 0/1/Y, L)$$

$$\delta(q_2, X) = (q_0, X, R)$$

$$\delta(q_v, Y) = (q_v, Y, R)$$

$$\delta(q_v, B) = (q_a, B, S)$$

$$\delta(q_v, 0/1) = (q_x, 0/1, S)$$